

SHUBHRA VARSHNEY

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LeetCode | GitHub | Codeforces

Education

Jaypee Institute of Information Technology

Bachelor of Technology in Computer Science and Engineering
CGPA: 9.3/10

Sector-62 Noida, India

2023 – 2027

City Vocational Public School

CBSE (Class XII) : 98.8%
CBSE (Class X) : 96.6%

Meerut, India

2021 – 2022

2019 – 2020

Skills

Languages

C, C++, Python, JavaScript

Backend & APIs

Node.js, Express.js, FastAPI, REST APIs, JWT Authentication

Frontend

React.js, HTML, CSS

Databases

MongoDB, MySQL

AI/ML

Machine Learning, Generative AI, LLM Integration, Model Evaluation

Concepts

Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems

Tools

Git, GitHub, Postman, Streamlit

Projects

ResQNet

GitHub | Live | MERN, Socket.io, React Leaflet, MongoDB, JWT

- Built a real-time disaster response platform using Socket.io to instantly broadcast citizen-reported emergencies to the command-center dashboard and assigned responders based on live GPS location.
- Engineered an automatic priority-scoring system and implemented Dijkstra's Algorithm from scratch for optimal responder routing, and applied the Haversine formula for nearest responder and shelter selection.
- Designed role-based access control (Citizen, Responder, Admin) using JWT authentication and developed interactive React Leaflet dashboards for live emergency monitoring, mission tracking, and shelter management.

AI Document Intelligence System

GitHub | FastAPI, React.js, Gemini API, Edge-TTS

- Built a full-stack AI platform using FastAPI, React.js, and Gemini API to process PDF documents and generate summaries, context-aware Q&A responses, and structured study, lecture, and lab planners.
- Developed a multi-speaker podcast generation pipeline using Gemini API, Edge-TTS, and Pydub to transform PDF document content into 2–10 minute conversational audio through script generation, voice synthesis, and audio merging.

Smart Metro Management System

GitHub | C++, Graph Algorithms, Data Structures

- Modeled a 25-station weighted metro network in C++ using object-oriented design with user/staff modes, smart-card fare handling, and multi-station facility management.
- Implemented Dijkstra's Algorithm for shortest-path routing and the Chinese Postman Problem for patrol route optimization over the graph-based metro network.

Achievements

- LeetCode Contest Rating: 1777** | Solved **600+** Data Structures & Algorithms problems on LeetCode.
- Achieved **Elite (Top 1%)** distinction in the **NPTEL Cloud Computing** certification (IIT Kharagpur) with a score of **94%**.
- GirlScript Summer of Code (GSSoC) 2025** – Open Source Contributor; contributed bug fixes and feature enhancements to the *WebDevIn100_Days* repository through GitHub pull requests.
- Smart India Hackathon (SIH) 2024** – Internal round qualifier with a healthcare system solution.

Leadership & Mentorship

IEEE Student Branch — Technical CSE Volunteer

- Mentored 150+ students in C++ workshop and assisted in organising IEEE technical workshops and coding events.